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The Clinical AI Safety Gap from Correct Answers to Safer Decisions

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Abstract

High accuracy alone does not establish that language-model assistance improves clinical safety. Drawing on 31 selected publications, this critical Perspective outlines an error-pathway ledger connecting incorrect outputs to display, confirmed viewing, clinician recognition, final decisions, and potential harm. It specifies distinct descriptive indicators and a randomized comparison for evaluating unsafe decisions. The framework provides an operational research design for prospective validation against established clinical evaluation approaches.

Keywords: large language models; clinical decision support; patient safety; human–AI collaboration; error detection; pragmatic evaluation.

1 Introduction

A correct response to a simulated clinical case is not equivalent to a safe decision in practice. The deployed intervention includes the model, the interface, the clinician's interpretation of the advice, workflow interruptions, and the ability to challenge incorrect recommendations. The appropriate question is therefore not simply "Can the model answer?" but "Does its use reduce harmful decisions compared with a credible alternative?" [1,10,17,18,22].

The distinction is consequential. Clinicians may perform better with an LLM on certain tasks, yet neither improvement over unaided clinicians nor the presence of human supervision establishes superiority to the best alternative, fewer unsafe decisions, or better patient outcomes [1,2,11]. Existing guidance already covers early clinical evaluation, lifecycle validation, and safety risks [26-30]. This article focuses more narrowly on the *measurable clinical passage of an erroneous recommendation* through an assisted decision, and on what a rigorous study would need to record to test whether that error is contained.

2 Approach and boundaries of evidence

2.1 Evidence selection and analytical approach

The analysis draws on 31 selected publications: 25 sources identified during exploratory work and six additional studies addressing recent advances in safety, evaluation, and reporting. Sources were chosen for relevance to clinician-facing decisions, incorrect advice, clinical deployment, and measurement methods. We distinguish evidence syntheses, controlled studies, conceptual frameworks, and reporting statements. Bibliographic records and selected central findings were checked against accessible publisher or indexed material. The synthesis is narrative and interpretive; full-text extraction and formal study-level quality appraisal were not carried out for every

source. Evidence maps inform the discussion of the literature [4,5,6], while guidance documents are interpreted according to their defined scope [24,25,28-30].

Accordingly, this paper is a targeted critical Perspective rather than a systematic review or a new meta-analysis. The selection does not establish exhaustive coverage of the literature. Numerical findings are attributed to their original publications and are not re-estimated or pooled across reviews.

2.2 Five levels of increasingly demanding clinical claims

The central methodological error to avoid is shifting from one outcome to another without a new comparison. We distinguish five levels of evidence rather than regarding them as interchangeable.

Table 1: Clinical claims and their necessary comparisons

Claim	Minimum target outcome / comparator	What it cannot alone establish
Model correctness	Adjudicated outputs on a representative case set	Clinician benefit or patient safety
Assistance benefit	Clinician+AI versus clinician without that AI	Superiority to the best standalone option
Team synergy	Clinician+AI versus clinician-only *and* AI-only	Fewer harmful decisions under routine conditions
Error containment	Wrong advice encountered, detected, corrected, and acted upon	Causal prevention of harm without a suitable control
Clinical net benefit	Pragmatic comparison, adjudicated severity, patient-relevant outcomes	Long-term transfer without external follow-up

2.3 No double counting across syntheses

Wang, Chen, Takita, Shan, and other reviews address overlapping domains but use different populations, inclusion periods, units of analysis, and outcomes [1,3,5,10,15]. Their review-level summaries are *not* independent patient cohorts. They cannot legitimately be added together to claim a novel sample size or combined into an overall "AI safety effect." We therefore emphasize contrasts in questions and designs, not numerical pooling.

3 Evidence that performance is context-dependent

A systematic review by Wang and colleagues pooled only two studies for each of several clinical performance outcomes; its estimates suggest task-dependent promise but substantial imprecision [1]. Task scope differs across oncology [12], medicines safety [13], patient advice [14], diagnostic comparisons [15], and retrieval augmentation [16]; controlled clinical reasoning studies are likewise workflow-specific [17-22]. This means that the intervention being assessed must be defined as a specific, versioned clinician–model workflow rather than a general technology category.

4 What the published quantitative evidence establishes

4.1 Performance with and without clinicians

Wang et al. analyzed ten peer-reviewed studies. For composite diagnostic and management performance, the random-effects mean difference favored clinician+LLM by 4.88 percentage points (95% confidence interval [CI] 0.65–9.12; $k = 2$). The 95% prediction interval was wide (-31.65 to 41.42), and the pooled time difference was 0.40 minutes (95% CI -4.18 to 4.97; $k = 3$). Their pooled binary correctness ratio was 1.59 (95% CI 0.08–32.74; $k = 2$), illustrating severe imprecision [1].

Takita et al. reviewed 83 diagnostic performance studies and reported pooled generative-AI accuracy of approximately 52.1%; the finding does not estimate correctness for a particular hospital or clinical workflow [10]. In an interdisciplinary systematic review, Vaccaro et al. synthesized 106

experiments and 370 effect sizes: human–AI systems underperformed the better independent agent on average (Hedges' $g = -0.23$, 95% CI -0.39 to -0.07) [11]. These studies address different estimands; they are not contradictory simply because their numerical signs differ.

Table 2: Selected published results; each denominator belongs to its source

Study	Reported result	Interpretive boundary
Wang et al., 2026 [1]	+4.88 percentage points; composite score; $k = 2$	Wide prediction interval, few combinable studies
Wang et al., 2026 [1]	+0.40 minutes (95% CI -4.18 to 4.97); $k = 3$	No overall time advantage established
Vaccaro et al., 2024 [11]	$g = -0.23$ versus best solo agent; 106 experiments	Interdisciplinary; not a medical-only estimate
Takita et al., 2025 [10]	52.1% pooled accuracy; 83 studies	Benchmark/case performance, heterogeneous tasks
Chen et al., 2026 [5]	4,609 publications; model-estimated 1,048 with patient data (95% CI 847–1,252); 19 prospective RCTs	LLM-assisted evidence mapping; estimates not direct census totals

4.2 What individual trials contribute

Randomized studies by Goh et al. and clinical reader studies by McDuff and Wu demonstrate that specific clinician–LLM combinations can alter diagnostic or management performance [17-20]. A perioperative randomized crossover study shows why claims must be tied to the evaluated workflow and participant characteristics [21]. Meta-analyses of clinician–LLM diagnosis and triage also face outcome and comparator heterogeneity [3,15]. These findings motivate task-specific testing, not an assertion that LLMs improve all services. Studies of rare diseases, oncology, multimodal prediction and retrieval augmentation target different populations and require distinct reference standards [7-9,12,16].

5 The central safety gap: wrong advice that survives

5.1 Human review is an observable process

Zeng, Lei, and Li systematically reviewed clinician responses to erroneous automated clinical advice. Their findings show that human oversight does not reliably intercept all incorrect outputs [2]. Crucially, this synthesis extends to *broader clinical automation*, including systems that are not LLMs; it must not be presented as an LLM-specific failure probability. A rigorous clinical LLM study should record both the *availability of an error for inspection* and what the clinician ultimately did with it. The mere presence of a clinician in the loop is a study condition, not a safety endpoint.

Wang et al. described factual errors in selected LLM-assisted documentation studies (roughly 26–36%, with differing error definitions) even where measured documentation quality improved [1]. A readable note can therefore contain incorrect clinical facts; comprehensibility and factual validity should remain distinct endpoints. Medication-related harm, patient-facing chatbot advice, and cancer-management evaluations expose additional failure modes such as unsupported recommendations and omitted context [12-14].

5.2 Real-world clinical evidence is scarce relative to publication volume

The LLM-assisted synthesis by Chen et al. identified 4,609 papers published from January 2022 to September 2025. Their Bayesian classification estimated 1,048 studies involving real-world patient data (95% CI 847–1,252), with 19 confirmed prospective randomized trials in that category [5]. These are the review authors' classifications, not original counts assembled for this manuscript; use of patient data alone does not imply a prospective live deployment. Vignettes and knowledge

tests cannot reproduce the interruption, escalation, uncertainty, and information quality of routine care [23].

5.3 Incomplete reporting weakens safety inference

Chen et al.'s 2026 *BMJ* review found 68 LLM trial reports, 15 protocols, and 378 registrations. Mean reporting completeness was 40% under TRIPOD-LLM across reports and 44% under CONSORT 2025 among randomized reports; the CONSORT-AI figure was 59% in the eligible subset [6]. Reporting gaps concerned model identity and version, prompt/configuration, input handling, data provenance, and harms documentation. A technology's uncertainty is difficult to investigate when the intervention cannot be reconstructed.

> **THE SAFETY QUESTION**
>
> Did an incorrect model suggestion reach a clinician, was it recognized, was the final decision changed, and did the resulting choice create a clinically consequential error? Each transition needs its *own* record and denominator; neither benchmark accuracy nor the label "human oversight" supplies those data.

6 What our proposal adds to existing frameworks

The proposed contribution is *not* a claim that patient safety, human factors, or lifecycle evaluation are new concepts. These are extensively addressed by existing literature. Clusmann et al. map LLM safety and security threats across the development and deployment lifecycle [26]; Ye et al. present a five-phase medical AI evaluation framework spanning technical validation, interaction, clinical evidence, and post-deployment monitoring [27]. DECIDE-AI addresses early-stage live evaluations [28]; TRIPOD-LLM provides LLM-specific reporting guidance [29]; CONSORT-AI covers AI intervention trials [30]. TRIPOD+AI covers prediction-model reporting and is complementary rather than a substitute for clinical trial guidance [24]. These contributions are essential and are not replaced here. A recent pilot-tested OMNI rubric evaluates the quality and safety of generative-AI outputs [31]; unlike that output-centered rubric, the proposed ledger targets linked actions through a final clinical decision. This distinction is a testable scope difference, not established superiority.

Table 3: Positioning relative to related 2022–2026 literature

Work	Established contribution	Specific complement proposed here
DECIDE-AI, 2022 [28]	Reporting early-stage live clinical AI evaluation	Prospective exposure/error/final-decision event ledger
Clusmann et al., 2026 [26]	Lifecycle threats, safety layers, responsibilities	Decision-level causal comparison and error survival
Ye et al., 2026 [27]	Five-phase evaluation and dynamic phase gating	Within-workflow incorrect-advice transition outcomes
Zeng et al., 2026 [2]	Meta-analysis of recognition/resistance to erroneous AI	Versioned, LLM-specific validation plan with clinical endpoints
TRIPOD-LLM / CONSORT-AI [29,30]	Standardized methodological reporting	Predefined minimal data fields for safety estimands
OMNI rubric, 2026 [31]	Pilot-tested ratings of generative-AI output quality/safety	Linking reviewed outputs to observed decisions

6.1 The error-pathway ledger as a testable addition

For each clinician–case *decision opportunity*, the proposed ledger links model version and output correctness, display, confirmed viewing, explicit recognition, any revision, final action, and adjudicated clinical risk. Never displayed, not viewed, and unknown states are kept separate. Its measurement performance will be assessed in a future validation study.

6.2 Why linked events matter

The key benefit is traceability. An incorrect output that is never shown is not equivalent to incorrect advice that is read, relied upon, then corrected; neither is equivalent to an unsafe final action. Reporting such cases as the same undifferentiated "hallucination rate" loses information that is central to clinical governance.

6.3 What would demonstrate genuine methodological novelty?

A positive validation study would show that independent assessors can apply the ledger reproducibly, that its error-transition outcomes distinguish assistance interfaces missed by conventional accuracy summaries, and that these measures correlate prospectively with independently adjudicated consequential errors. These criteria define prospective hypotheses for empirical testing.

7 The proposed error-pathway measurement architecture

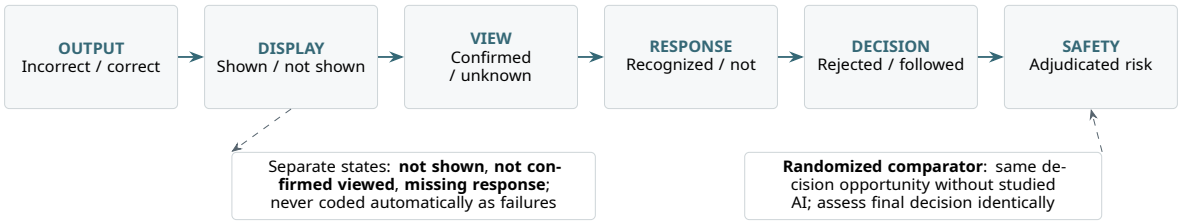


Figure 1: Event linkage proposed for research, not a causal chain inferred from observational transitions. Branches permit absent display, uncertain viewing, and missing responses.

7.1 Separate denominators and observable event states

For one prespecified index recommendation per decision opportunity, let N be all randomized eligible encounters in the AI-assisted arm, E those with an adjudicated incorrect output, $D \subseteq E$ errors displayed, $V \subseteq D$ errors *confirmed viewed* (using a preregistered operational rule), $R \subseteq V$ errors explicitly identified by a clinician as wrong, and $K \subseteq V$ errors rejected in the final recorded choice. Let $U \subseteq R \cap K$ represent rejection with documented recognition preceding the decision. An unread display must not be counted as confirmed viewing; ambiguous viewing is *unknown*, not "not viewed." Logging and self-report are different evidence sources, and confirmation of a click is not proof of comprehension.

Table 4: Proposed research indicators with distinct denominators

Indicator	Numerator / definition	Denominator
Incorrect-output proportion	Adjudicated incorrect index outputs (E)	Evaluable index outputs
Display of incorrect advice	Incorrect recommendations shown (D)	All wrong outputs (E)
Confirmed-view proportion	Wrong advice confirmed viewed (V)	Displayed wrong advice (D)
Documented recognition	Explicitly recognized wrong advice (R)	Confirmed viewed errors (V)
Final rejection after viewing	Wrong advice rejected (K)	Confirmed viewed errors (V)
Recognition-linked correction	Rejection preceded by recognition (U)	Recognized errors (R)
Clinically unsafe final choice	Unsafe choices by blinded adjudication	All randomized opportunities (per arm)

Thus $|R|/|V|$ is *documented recognition among confirmed viewed wrong outputs*, not a latent cognitive recognition probability; $|K|/|V|$ measures observed final rejection, not necessarily recognition-driven correction. Unknown viewing and missing outcomes require their own counts, sensitivity

analyses, and transparent reporting. When multiple outputs occur within one encounter, their nesting and clustering must be explicitly modeled; no simple addition of event denominators is justified.

7.2 Clinical error grading and patient harm are distinct

Before any prospective study, a multidisciplinary panel should approve an explicit reference standard and classify final choices as *appropriate*, *uncertain/indeterminate*, or *inappropriate*, then assess prespecified harm potential (none, minor, moderate, severe, life-threatening) using case-context criteria. For the proposed primary endpoint, a *clinically consequentially unsafe choice* is an independently judged inappropriate action with credible potential for at least moderate harm if implemented. Two clinicians blinded to allocation and AI text should adjudicate independently with documented resolution of disagreements; agreement must be reported. *Realized patient harm*, if studied, requires its own follow-up and denominator and must not be inferred from hypothetical risk labels.

7.3 Randomization identifies a different quantity

The causal deployment estimand is the intention-to-treat risk difference in unsafe final choices across randomized workflows, not a conditional error-path transition. Display, viewing, and recognition occur after assignment; comparing only exposed or incorrect-output encounters could induce selection bias. The ledger's transition indicators are descriptive and should not be called the proportion of harm caused or prevented by the model.

8 A prospective validation and analysis blueprint

8.1 Study design and comparators

The proposed validation sequence comprises: (1) stakeholder and clinician consensus on safety criteria, followed by a pilot to estimate event prevalence, rater agreement, and feasible capture; (2) preregistered external measurement validation and a multi-reader randomized experiment with clinician-only versus clinician+LLM (AI-only benchmarking as context); and (3) a multicenter pragmatic comparison with prospective monitoring. The sequence is a study design for future implementation. Simulated experiments may ethically expose consenting clinicians to deliberately erroneous advice using synthetic or deidentified cases, but such advice must not reach actual patient care without independent safeguards. Live studies should use approved safeguards and escalation rules [27,28,30].

The *primary pragmatic endpoint* should be the proportion of all randomized decision opportunities yielding an independently adjudicated clinically consequential unsafe final choice, using the same rubric in both arms. Secondary outcomes include correct decisions, severity of unsafe actions, time to a final decision, false alarms, user burden, and (where observable and adequately powered) patient-level outcomes. Analyses should preserve clinician, case, and site clustering; report absolute risk differences alongside relative effects and uncertainty; and account for crossover, repeated cases, or model changes. Report safety rates separately by risk level, with explicit justification for any composite scoring.

Table 5: Minimum record for a reproducible safety evaluation

Field group	Required traceable information
Decision context	Deidentified case ID; site; specialty; risk category; target decision
AI intervention	Provider/model/version; retrieval sources; prompt policy; interface; timestamp
Adjudication	Reference standard; blinded raters; disagreement and severity rules
Exposure and response	Output correctness; viewed/not viewed; clinician recognition; revision; override
Clinical endpoint	Final decision; error category; severity; outcome and follow-up window
Protocol and analysis	Allocation; unit of analysis; estimand; missing data; monitoring rules

8.2 Estimands and falsifiable validation targets

The primary causal estimand in a randomized evaluation is the intention-to-treat risk difference $P(Y{=}1 \mid Z{=}1) - P(Y{=}1 \mid Z{=}0)$, where Y is a prespecified consequentially unsafe final decision and Z is assignment to the AI-assisted workflow. The ledger's conditional transition rates are *descriptive mechanisms*, not substitutes for this comparison. A credible protocol should preregister: (i) weighted agreement and exact adjudicator disagreement for error/severity labels; (ii) proportions of linkable and indeterminate exposure/response records; (iii) incremental discrimination and calibration versus simpler model-accuracy summaries, assessed in separate validation cases and sites; and (iv) time and resources required to apply the ledger. *Working success thresholds* for completeness, reliability, and minimally important predictive gain should be agreed with clinicians and patient representatives *after feasibility work but before the main validation sample is examined*; choosing them afterward would invalidate a confirmatory claim. Failure to satisfy them would refute practical added value. These parameters would be estimated in the planned feasibility and validation stages.

8.3 Power and missing-data planning

Sample size for the randomized primary endpoint must be calculated using pilot-estimated baseline unsafe-decision risk p_0, a stakeholder-defined minimum worthwhile absolute difference Δ, type-I error (for example, two-sided $\alpha = 0.05$), and desired power (for example, 80–90%). The calculation must allow for clinicians handling multiple cases, clustering by clinician/site (including an estimated intracluster correlation), rare severe events, and incomplete ascertainment. For a simple independent two-arm planning approximation, $p_1 = p_0 - \Delta$ and

$$n_{\text{arm}} \approx \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \left[p_0(1-p_0) + p_1(1-p_1)\right]}{(p_0 - p_1)^2},$$

with subsequent clustering and loss inflation. The expression is a planning approximation; sample-size estimates require empirical inputs. The primary model would estimate an arm-level marginal absolute risk difference, using clustered/multilevel methods and prespecified handling of unknown or missing outcome adjudication. Reporting shall include confidence intervals, sensitivity analyses, harms monitoring, and subgroup equity analyses where feasible.

8.4 Failure modes and checks

The ledger itself may fail if displayed material is mistaken for read advice, if clinicians do not report recognition, if observation changes behavior, if the reference answer is disputable, or if consequential errors are too infrequent to estimate precisely. Address these by automatically recording interface interactions where lawful, applying a blinded panel with predefined disagreement procedures, quantifying missingness, and assessing inter-rater agreement. A randomized study can estimate the effect of *assignment to an AI-supported workflow*; it does not establish that every unsafe choice following incorrect advice was caused by that output. Version drift, data privacy, subgroup disparities, and monitoring burden should be evaluated rather than assumed away [6,23,26,27,29].

8.5 Reporting and registration

A subsequent systematic review would need an advance protocol, database-specific reproducible searches, duplicate screening, and study-level risk-of-bias assessment; PRISMA 2020 specifies its reporting [25]. A clinical trial should instead follow the applicable CONSORT guidance and AI extension [30]; early-stage live evaluations may draw on DECIDE-AI [28], and LLM-specific reporting should address the TRIPOD-LLM items relevant to the study design [29]. These are reporting aids, not substitutes for sound research methods.

9 Interpretation, limitations and implications

9.1 What changes with an error-pathway focus?

The problem of clinical LLM deployment is not solved by making a model produce a higher fraction of reference-matched answers. Because advice is filtered through human judgment, performance depends on when the model is consulted, whether clinicians identify its incorrect claims, and whether an error survives into a consequential clinical choice. The proposed ledger separates output error, verified exposure, documented recognition, final rejection, and independently adjudicated unsafe decisions. It also allows an interface with similar model accuracy to be compared on the basis of whether clinicians can correct its errors.

This proposal neither displaces broad safety taxonomies [26] nor competes with lifecycle evaluation schemes [27]. Its intended additional value is a decision-level measurement specification that can be tested inside those frameworks. The correct ultimate comparator is a clinically viable alternative workflow, with patient benefit and burden assessed jointly. Depending on the task, this might be conventional decision support rather than a fully generative model; standalone LLM accuracy should not be a presumed clinical benchmark [11,22].

9.2 Strengths and limitations

The paper differentiates source populations, contrasts established frameworks, and defines measurable variables for clinical evaluation. Its purposive evidence selection may omit relevant contrary findings, and the proposed ledger has not yet been tested for construct validity, reproducibility, feasibility, or incremental utility. A prospective protocol, independent clinical adjudication, patient and stakeholder participation, and comparison with existing measures are required to determine its scientific and practical value. Evidence from broader automation and interdisciplinary human–AI research provides context rather than LLM-specific clinical treatment-effect estimates.

10 Conclusion

Clinical AI safety asks a more demanding question than whether an LLM is often correct: *what happens when it is wrong?* An error-pathway approach allows incorrect recommendations to be traced through viewing, recognition, action, and independently judged risk. The proposed measurement design offers a basis for comparing workflows using patient-relevant outcomes. Its practical value will depend on prospective testing of reliability, feasibility, and the additional information provided beyond conventional accuracy summaries.

Statements and declarations

Data availability. This critical Perspective did not generate primary patient data or original statistical estimates. The underlying publications are cited in the references.

Ethics. This work analyzes previously published literature and involves neither direct participant recruitment nor identifiable patient data.

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